

CSE400 – Fundamentals of Probability in Computing

Lecture 7: Expectation, CDFs, PDFs and Problem Solving

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1. The Cumulative Distribution Function (CDF)

1.1 Definition

Let X be a random variable. The **Cumulative Distribution Function (CDF)** of X is defined as

$$F_X(x) = P(X \leq x).$$

Step-by-Step Reasoning

1. Consider the event $\{X \leq x\}$.
2. This event contains all outcomes for which the random variable takes a value less than or equal to x .
3. The probability of this event defines the cumulative distribution up to the point x .

This definition applies to both discrete and continuous random variables.

1.2 Properties of the CDF

From the definition $F_X(x) = P(X \leq x)$, the following properties follow directly:

1. Non-decreasing property

If $x_1 < x_2$, then $\{X \leq x_1\} \subseteq \{X \leq x_2\}$, and hence

$$F_X(x_1) \leq F_X(x_2).$$

2. Boundary conditions

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow +\infty} F_X(x) = 1.$$

3. Right-continuity

The CDF is right-continuous due to properties of probability measures.

4. Interval probability For $a < b$,

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

1.3 Example (as per slide structure)

An example CDF is obtained by accumulating probabilities for a discrete random variable or by integrating a density function for a continuous random variable up to the value x . No additional numerical example is introduced.

2. The Probability Density Function (PDF)

2.1 Definition

For a **continuous random variable** X , the **Probability Density Function (PDF)** $f_X(x)$ is defined through its relationship with the CDF:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

Where differentiable,

$$f_X(x) = \frac{d}{dx} F_X(x).$$

Step-by-Step Reasoning

1. The CDF accumulates probability up to x .
2. The PDF represents the rate of change of this accumulated probability.
3. Differentiation of the CDF yields the PDF.

2.2 Properties of the PDF

1. **Non-negativity**

$$f_X(x) \geq 0 \quad \text{for all } x.$$

2. **Total probability equals one**

$$\int_{-\infty}^{+\infty} f_X(x) dx = 1.$$

3. **Probability over an interval**

For $a < b$,

$$P(a < X < b) = \int_a^b f_X(x) dx.$$

2.3 Example (as per slide structure)

Given a PDF, probabilities are computed by integrating over the desired interval, and the CDF is obtained by integrating from $-\infty$ to x .

3. Expectation of Random Variables

3.1 Definition

The **Expectation** of a random variable X is defined as:

- **Discrete case**

$$\mathbb{E}[X] = \sum_x x P(X = x).$$

- **Continuous case**

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) dx.$$

Step-by-Step Reasoning

1. Each value of X is weighted by its probability or density.
2. The sum or integral aggregates these weighted values.
3. The result represents the long-run average value.

3.2 Example (as per slide structure)

Expectation is computed directly using the PMF or PDF without introducing additional numerical examples.

4. Expectation of a Function of a Random Variable

Let $Y = g(X)$. The expectation of Y is:

- **Discrete case**

$$\mathbb{E}[g(X)] = \sum_x g(x) P(X = x).$$

- **Continuous case**

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx.$$

The expectation is computed directly with respect to the distribution of X .

5. Linear Operation with Expectation

5.1 Linearity Property

For random variables X and Y , and constants a and b :

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Justification

1. Apply the definition of expectation.
2. Use linearity of summation or integration.
3. Factor out constants.

No independence assumption is required.

6. Moments and Central Moments

6.1 nth Moment

The **nth moment** of a random variable X is

$$\mathbb{E}[X^n].$$

6.2 Central Moments

The **nth central moment** is

$$\mathbb{E}[(X - \mu)^n], \quad \mu = \mathbb{E}[X].$$

6.3 Variance

Variance is the second central moment:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

Using expansion,

$$(X - \mu)^2 = X^2 - 2\mu X + \mu^2,$$

which gives

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

6.4 Skewness

Skewness corresponds to the third central moment and measures distribution asymmetry.

6.5 Kurtosis

Kurtosis corresponds to the fourth central moment and measures tail behavior.

End of Lecture 7 Scribe